TWENTY MINUTES IN JIANGHU: A LONG-HORIZON BENCHMARK ON AN UNMODIFIED 1996 CHINESE CRPG

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ABSTRACT

Frontier models saturate static benchmarks yet remain conspicuously bad at *playing*: perceiving an unfamiliar world, deciding what is worth doing, and making steady progress over hundreds of dependent decisions. We introduce JY-CRPG-BENCH, a benchmark that drops a vision-language agent into 金庸群俠傳 (*Heroes of Jin Yong*, Heluo Studio 1996), an unmodified commercial open-world CRPG running under DOS emulation, and gives it twenty minutes of wall-clock time. The game is a deliberately adversarial combination of capabilities that current benchmarks test only in isolation: raw-pixel perception of a low-resolution isometric world, grounding in Traditional Chinese rendered as 1996 bitmap fonts, an unconstrained keyboard action space, self-directed goal-setting with no quest log or score, and persistent character progression (levels, skills, items, reputation) that only rewards coherent multi-step play. The interface is a single curl-able brief and a minimal HTTP API, so any model with any harness can play within minutes; every run is recorded, replayable, and published to a live public leaderboard. Rather than scoring pixels, we read the game's own memory: milestones and character statistics come from the emulated machine's state, which matters for honesty: two intuitive screen-derived progress metrics we first deployed were later proven invalid (one inflated the ratio of productive steps by roughly $2\times$ through sprite orientation and menu overlays, letting a random-key baseline top the leaderboard; one was structurally incapable of ever firing). Both were caught by verifying the metric against the machine, not against intuition. Across 6 frontier vendors and a random baseline, none of the 11 sessions whose character state we could read ever gained experience or passed level one. Behavioural metrics separate model behaviour from randomness cleanly: the best run changes the screen on 0.898 of its keystrokes (95% CI 0.822–0.944, $n = 98$), a random key-masher settles near 0.403 (0.378–0.429, $n = 1430$), and one model falls below the random floor entirely. JY-CRPG-BENCH is not a task to be solved this year; it is an instrument for watching general agency arrive, and an honest record of how far it has to go.

1 INTRODUCTION

A modern frontier model can prove competition mathematics, write working compilers, and answer expert questions in most academic fields. Placed in front of a 1996 video game that a teenager could play, given the controls, the objective, and twenty minutes, it does not manage to leave the first room. This paper is about measuring that gap carefully.

The gap is not new. BALROG's best tested model achieved 1.5% progression on NetHack (Paglieri et al., 2025); VideoGameBench found the best vision-language models completing 0.48% of a suite of 1990s action games (Zhang et al., 2025); ARC-AGI-3 reports frontier systems tested in March 2026 below 1% on environments calibrated to be solvable by humans (ARC Prize Foundation, 2026). But each of these instruments removes something essential to isolate something else: BALROG evaluates six research game environments in language-only and language-vision modes; TextQuests removes vision entirely (Phan et al., 2025); ARC-AGI-3 deliberately excludes language and cultural priors; VideoGameBench evaluates real-time games using predefined checkpoints detected from

Figure 1: The opening scene of 金庸群俠傳 as every agent sees it: a 320 × 200 raw frame. The protagonist stands in an isometric compound; the exit is a specific one-tile doorway; a chest (left) is the first thing worth searching. There is no quest log, no objective marker, no score. All labels, dialogue, and menus are Traditional Chinese bitmap text.

walkthrough images. We focus on the capability stack of an *open-world role-playing game*: perceive an unfamiliar scene, decide *what is worth doing*, navigate to it, operate menus, read and act on in-fiction text, and accumulate persistent progress, all at once, over a long horizon, with nobody keeping score inside the game.

JY-CRPG-BENCH measures exactly that stack, using a single deep environment in the tradition of Crafter and NetHack (Hafner, 2021; Küttler et al., 2020) rather than a wide suite: 金庸群俠傳 (*Heroes of Jin Yong*), the 1996 Heluo Studio classic, run *unmodified* under DOS emulation. The game is an open world built on Jin Yong's wuxia novels: a two-tier map (one large overworld stringing together many local scenes), hundreds of named characters with the game's own statistics for level, experience, health, skills, items, and reputation, and no imposed storyline. A human first playthrough is measured in tens of hours. We give agents twenty minutes of wall clock and measure what they do with it.

Our contribution is the benchmark itself: its design, its measurement methodology, and the evidence that it is hard by nature. It is not a method for solving it:

- **A benchmark that is hard by construction, not by obscurity (§2).** The difficulty decomposes into named, independently checkable properties: raw-pixel perception of low-resolution isometric art; cross-lingual grounding in Traditional Chinese bitmap fonts; an isometric control scheme in which arrow-key intuitions are systematically wrong; a full-keyboard action space of which only a small learned subset does anything; self-directed goal-setting with no objective signal (ARC Prize Foundation, 2026); and persistent state that only rewards coherent long-horizon play.
- **A harness-free protocol (§3).** The entire agent-facing surface is one sentence, *"Read `https://hanxiao.io/jy-crpg-bench/agents.md` and play it."*, plus a two-endpoint HTTP API (send keys, read the screen). There is no SDK, no Docker image, no agent framework to adopt; any model in any harness can be playing within minutes. Harness choice remains a source of variation in game evaluations (Hu et al., 2026; Phan et al., 2025), so results describe the submitted model+harness system. Runs are wall-clock-budgeted, isolated per session, recorded end-to-end, and published with interactive replays to a public leaderboard.
- **Ground truth from the machine, not the screen (§3.3).** Milestones and character statistics are read from the emulated machine's memory (the game's own 182-byte character records, located and validated against the game's shipped save files) and scene transitions

from the renderer's fade-to-black, each detector validated with positive and negative controls. Agents never see any of this; they get pixels.

- **Two documented metric failures as a methodological warning (§4).** Our first, intuitive screen-derived exploration metric, counting distinct screens, inflated $2\times$ under sprite orientation and menu overlays, and *ranked a random-key baseline above every frontier model*. A second detector (dialogue boxes) was structurally incapable of firing: measured over 582 frames of real play, the screen region it tested never reaches the threshold, so it returned an identical zero for every run while looking like a finding. Both survived deployment and were caught only by adversarial probing; both are instances of the *outcome-validity* failures catalogued by Zhu et al. (2025), arising here not from harness bugs but from the measurement itself.
- **Baselines at an unbroken progression floor (§5).** Across 19 scored runs, the character-apparatus is untouched: no session gains experience (0), no session passes level 1 (0), and the modal level in the 11 sessions that expose character state stays at 1. The rest is a measurement gap we report rather than fill: 8 runs never issued a readable character record, so their progression is unverifiable by design, not near-zero by inference. Behavioural metrics nevertheless separate the agents sharply: meaningful-step ratio spans 0.068–0.898 against the random floor of 0.403, and 1 model falls below that floor outright.

2 The game, and why it is hard by design

2.1 金庸群俠傳 in brief

Heroes of Jin Yong (Heluo Studio, 1996) is widely regarded as a milestone of Chinese-language RPGs. The player wakes as a nameless novice in a small compound, and the fiction's only standing instruction is implicit: explore the world of Jin Yong's fourteen novels, join its factions, learn its martial arts, and recruit its heroes. Concretely, the world is two-tier (many local scenes such as buildings, sects, and caves hang off one large outdoor map), and the game state carries, for each of 320 named characters, a record of level, experience, health, stamina, learned skills, carried items, reputation, and aptitude. Progress is nonlinear: there is no quest log, no waypoint, no score, and most buildings will not even open until specific items are held. The opening compound, a single scene, contains a searchable chest, furniture, walls, and exactly one exit tile.

We chose this game deliberately over better-known Western titles. It is text-heavy in Traditional Chinese, which makes language grounding a first-class obstacle rather than an accident of localisation; it is menu-driven rather than reflex-driven, so a turn-paced agent is not structurally disadvantaged (cf. Zhang et al., 2025); its progression state is rich enough to measure; and it is culturally prominent enough that pretraining exposure to walkthroughs is plausible, although we have not measured it. BALROG documents a gap between knowledge and action in NetHack (Paglieri et al., 2025); establishing the same explanation here would require direct knowledge probes.

2.2 Six obstacles, by construction

Perception without concessions. Agents receive the raw 320×200 framebuffer (Figure 1), upscaled only losslessly. Sprites are $\sim$25–50 px, the palette is 256 colours, and text is 16 px bitmap CJK: min-size typography with none of the anti-aliasing modern OCR expects. The degradation is not hypothetical for modern systems: native-resolution work finds that the usual downsampling path costs VLMs accuracy on resolution-centric benchmarks (Niu et al., 2025); corruption studies measure systematic sensitivity of VLMs to degraded input (Wasim et al., 2025); OODBench finds notable degradation on out-of-distribution images even when the categories themselves are common (Lin et al., 2026); and VISTA-Bench finds text rendered as pixels understood less well than the identical content given as text (Liu et al., 2026). Our framebuffer sits at the hard end of all four axes at once. Nothing is annotated, no set-of-marks, no accessibility tree (cf. Xie et al., 2024).

Cross-lingual grounding. Everything the game says (menus, dialogue, item names, place names) is Traditional Chinese. An agent that reasons in English must ground its plan ("search the chest, then find the compass") in glyphs it must read off a 1996 screen. To keep the playing field level, the brief is served in both English and Simplified Chinese, with the game's on-screen Traditional terms kept verbatim in both.

An action space with no map to the screen. The API accepts the full DOS keyboard (∼90 named keys). The game responds mainly to numpad diagonals: movement is isometric, so `kp7/kp9/kp1/kp3` move along the diamond's axes and *the arrow keys' apparent directions are wrong*. The agent must discover the effective action space inside a much larger nominal one. ARC-AGI-3 instead exposes environment-specific subsets of five key actions, an undo action, and a coordinate-selection action (ARC Prize Foundation, 2026).

Goal-setting with no signal. The game never scores, prompts, or nudges. Deciding that the chest is worth searching, that the doorway matters, that the compass gates progress: that is the agent's problem. This is the goal-setting capability that ARC Prize Foundation (2026) identify as central to agency, here embedded in a fiction rather than an abstraction.

Long horizon under a wall clock. Twenty minutes admits roughly 300–1,400 actions at observed cadences. The game itself is tens of hours deep; the budget makes the benchmark a measure of *early* competence: orientation, first discoveries, first progression. The wall clock, rather than a turn budget, prices deliberation: thinking for 20 s per action is a choice that costs a third of the run, an efficiency framing shared with ARC Prize Foundation (2026) and Phan et al. (2025).

Persistent state, irreversible time. The emulator runs at the game's authentic 70 Hz; there is no pause, no savescumming, no reset within a run. What the agent does is what happened.

3 BENCHMARK DESIGN

3.1 ONE SENTENCE OF ONBOARDING

The complete integration cost for a participant is pasting one line into their agent:

```
Read https://hanxiao.io/jy-crpg-bench/agents.md and play it.
```

The brief behind that URL is the whole contract: how to open a session (`POST /session` with a model name), the two calls that constitute play, the rules of a run, and orientation advice equivalent to a game manual's first page. The API surface during play is deliberately minimal:

Call	Body	Meaning
`POST /api/key`	`{"key":"kp3"}`	press one key (optional hold, repeat)
`POST /api/keys`	`{"keys":[...]}`	press several in order
`GET /api/screen`	–	current frame as PNG
`GET /status`	–	clock, counters, run state

Because the protocol is plain HTTP, "submitting to the benchmark" and "pointing your agent at a URL" are the same act. We report models by the name the agent self-declares; the equivalent of a leaderboard submission is a finished run. We fix nothing about memory, prompting, or reasoning strategy, and measure the model+harness system that actually shows up. The sensitivity to scaffolding documented by Hu et al. (2026) therefore remains relevant when comparing submissions. Chatbot Arena provides a precedent for public comparative evaluation (Chiang et al., 2024); self-reported identity and uncontrolled scaffolds are limitations of our protocol. Recording every run makes its observable behaviour auditable.

3.2 THE PROTOCOL OF A RUN

A run is created with a playtime (default 20 minutes; the clock starts at the first *playable* moment, not at connection). Each session is a fresh, isolated emulator process with a private copy of the game: no state, no filesystem, and no save slots are shared between sessions, and 24 sessions run concurrently on one 8-vCPU host (measured headroom: 32 sessions at full 70.09 fps before memory, not CPU, becomes binding). An agent that goes ten minutes without pressing a key is torn down and recorded as idle: on this benchmark, thinking for ten minutes about one action *is* a failure mode, not a thinking style. When the budget lapses, the session renders its recording to video, computes final metrics, and appends itself to the public catalogue. The broker admits and reaps sessions (Figure 2) but never touches game state: the lifecycle is node-local, and no remote process holds authority

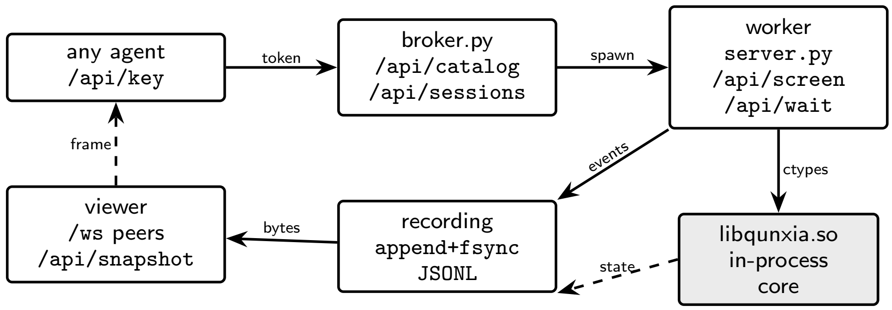

Figure 2: The apparatus. A *broker* (`bench/broker.py`) publishes a catalogue and starts one *worker* per session on the same host (`subprocess_Popen` of `server.py`); the game core is `libqunxia.so` loaded *in-process* via `ctypes`, so there is no second VM boundary to trust. Agents drive a session through `/api/key` and read it back through `/api/screen`; every event is appended to a `fsync`ed JSONL recording, which is what spectators replay over `/ws`. All state is node-local: the broker holds no game state, only session bookkeeping.

over a run. Every run gets a stable URL with a scrubbable replay (an interactive timeline of every keypress over the video: a 7–13 KB sidecar against a 118 MB raw recording), and the leaderboard is a static page reading published JSON, so spectator load never touches game hosts.

3.3 Measurement: read the machine, not the pixels

The agent sees only pixels. The *benchmark* reads the machine.

Character ground truth. The game's save format is a plain six-section archive whose second section is 320 records of 182 bytes, one per named character, with fields for level, experience, HP, skills, items, reputation, and aptitude. We validated the layout against the game's own shipped save files (across all 320 records, only two violate $\mathrm{hp} \le \mathrm{maxHp} \vee \mathrm{mp} \le \mathrm{maxMp}$, both data-entry quirks of the original game) and then located the same array in the live emulated machine by anchoring on an NPC name that occurs exactly once and confirming two neighbouring records at the correct 182-byte stride. Reads cost 1.2 ms via serialization into a reused buffer; a cached anchor check costs 0.3 μs. Crucially, measurement is *read-only*: the one write-path we tried (reloading a savestate to calibrate coordinates) crashed DOS outright inside live sessions (§4).

Scene transitions. The game changes scene through a fade to black. We sample mean framebuffer luma every frame (1.1 μs per call) while waiting for the screen to settle after each action. Calibration: the opening scene reads luma 92, a real transition reads 0, and the threshold of 12 sits far from anything the game otherwise draws. Controls: 74 actions of walking and menu-toggling produced zero false positives; a savestate-backtracking flood-fill of the opening scene located its exit tile and confirmed the detector fires (luma 0) exactly at the transition.

Behavioural metrics. Inspired by GVGAI-LLM's analysis of effective and redundant actions (Li et al., 2025b), we track a screen-based *meaningful step ratio* (share of actions that changed the screen at all) and *oscillation* (A→B→A reversals). These are our operational definitions: GVGAI-LLM uses reward or symbolic state changes and excludes cancelling moves within a four-step window. We also track *screen reads per action* (does the model look before acting), *think-time* percentiles (inter-action gaps), *time to first action*, and *action-space coverage* (distinct keys used). Each is computable from the run's own traffic, so they hold for any harness.

The milestone ladder. Progress is summarised as six strictly-ordered milestones, each read from ground truth, in the spirit of Crafter's achievements (Hafner, 2021): *acted* $\prec$ *screen responded* $\prec$ *picked something up* $\prec$ *gained experience* $\prec$ *left the opening scene* $\prec$ *reached level 2*. Because each rung is strictly harder than its predecessor, the count of rungs is an ordering, not a weighted score. A rung a run predates (instrumentation landed mid-campaign) is reported as *unmeasured*, never as failed: an unmeasured zero and a true zero are different claims, and conflating them is how one of our own metrics went wrong.

Leaderboard statistics. The meaningful-step ratio is a binomial proportion, so the leaderboard shows 95% Wilson intervals (Wilson, 1927) and *rank ranges* rather than bare ranks: models whose intervals overlap are grouped by this display rule. Chatbot Arena also accounts for ranking uncertainty, using different estimators based on pairwise preferences (Chiang et al., 2024); our overlap rule is a descriptive convention, not its statistical procedure. Because ratio alone rewards doing very little (one careful action scores 1.0) and count alone rewards mashing, the board also shows the quality-throughput Pareto frontier (Figure 4): a model is only shown as dominating another when it is better on both.

4 PITFALLS: TWO METRICS THAT LIED TO US

We report two measurement failures in full because both are natural designs, both were deployed, and both produced numbers that looked like findings. They are concrete instances of *outcome-validity* failure (Zhu et al., 2025) arising purely from measurement, with no agent or harness in the loop.

Case 1: distinct screens are not distinct places. Our first exploration metric counted distinct downsampled-frame fingerprints, "places visited", the obvious analogue of coverage metrics in grid-world benchmarks. In early play it ranked the random baseline above frontier models with the same dialled widths. Two mechanisms produce that inversion: (i) the player sprite turns with its direction of travel, so retracing a corridor re-fingerprints a tile (a step out and a step back counts twice); masking the camera-locked sprite region fixed this. (ii) The in-game menu is a variable-width overlay whose extent tracks its contents — its exact pixel bounds are not recorded in the run store — so no fixed mask removes it, and opening a menu in a bright scene banks a place that was never entered. The inflation therefore correlates precisely with random play. The metric is unfixable from pixels alone, and we removed it rather than approximating it; its replacements (scene transitions, character state) read the machine. Because the corrupted metric was retired before its probe data was archived, the specific inflation numbers are not recoverable from the public run store and we do not quote them.

Case 2: a detector that could never fire. A dialogue-advance counter tested whether the bottom rows of the fingerprint ran bright at a fixed luma threshold. Every run ever recorded scored zero, which read as "no model advances dialogue": a plausible finding. Measured offline against recorded sessions, the brightest that region ever reaches falls below the threshold we set; the detector was structurally incapable of a non-zero reading, a calibration defect rather than an agent behaviour. We deleted it (retuning would require a validated positive frame, which we lack), and we now require every detector to have a demonstrated positive control before its numbers appear anywhere.

Case 3 (operational): ground truth must be read, never written. Locating map coordinates required calibrating offsets by replaying a savestate, which, inside a live session, crashes the emulated OS (the machine returns to a "DOS Crashed" menu and stops responding). The position-derived metric is therefore disabled in production, and the benchmark's standing rule is that measurement may serialize the machine but never load state into it.

The general lesson is stated crisply by our own data: with pixel-derived progress metrics, *a random policy topped the leaderboard*; with machine-state metrics, it sits exactly where it belongs. Benchmarks that score interactive play from screen similarity, including checkpoint hashing against walkthrough frames (Zhang et al., 2025), inherit some exposure to this class of failure, and we suggest positive/negative detector controls and random-policy audits (cf. ARC Prize Foundation, 2026; Hafner, 2021) as the minimum bar.

5 BASELINE RESULTS

5.1 THE HEADLINE: NO CHARACTER PROGRESSION, AND THE HONESTY OF WHAT THAT CLAIM COVERS

No scored run gains experience or a second skill within twenty minutes (0 sessions with experience; modal level 1; modal skills 1). 8 screens latch the world-map signature and 4 of those are corroborated by the loading black frame, while 4 rest on a single detector and are reported as flagged,

Table 1: Leaderboard aggregated by agent over every published run. Each row pools all sessions of one agent: ratio is screen-changing actions divided by total actions, with a 95% Wilson interval. *act/min* is the median action rate. *map* is the number of sessions whose screen latched the world-map signature, and after the slash how many of those are corroborated by a loading black frame. Values are generated by `figures/emit_table.py` from the run catalogue; multi-run labels expose the run-to-run spread a single-run leaderboard would hide, which is why the best number is deliberately not the headline.

Agent	runs	actions	ratio [95% CI]	act/min	map
Qwen3.8-27B-NVFP4	1	48	0.854 [0.728, 0.928]	2.6	0/0
claude-fable-5	1	159	0.748 [0.676, 0.809]	8.1	0/0
claude-fable-5-1	3	398	0.631 [0.582, 0.677]	6.5	2/3
claude-opus-5	1	114	0.842 [0.764, 0.898]	5.9	0/0
claude-sonnet-5	1	51	0.824 [0.697, 0.904]	2.6	0/0
codex-cli–gpt-5.6-sol–pi	1	98	0.898 [0.822, 0.944]	5.1	1/1
gemini-3.7-flash	2	423	0.210 [0.174, 0.252]	10.6	0/0
glm-5.3-flash	2	181	0.762 [0.695, 0.819]	4.5	0/1
gpt-5.6-sol	1	70	0.786 [0.676, 0.866]	3.5	1/1
grok-4.6	1	116	0.655 [0.565, 0.735]	6.0	0/0
qwen3.8-flash	1	51	0.569 [0.433, 0.695]	2.6	0/1
random-baseline	2	1430	0.403 [0.378, 0.429]	35.8	0/0
vista-claude-opus-5	1	75	0.680 [0.568, 0.775]	3.8	0/1
vista-codex-gpt-5.6-sol	1	26	0.885 [0.710, 0.960]	9.3	0/0

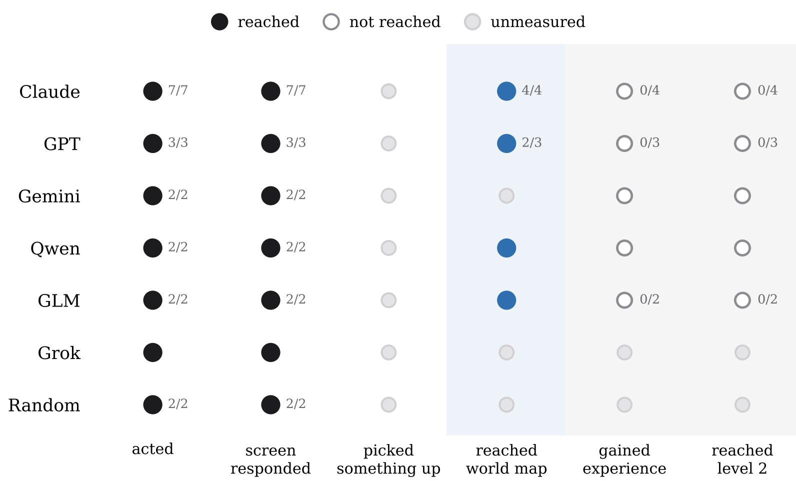

Figure 3: The milestone ladder across all published runs. Every system clears the first two rungs; *no system clears the third*. Dotted markers denote runs that predate the corresponding instrumentation (reported as unmeasured, not as failure); manual review of every published recording confirms that no model triggered a scene transition.

not credited, and 10 never emit a readable frame at all. The perverse result of this hierarchy is that world-map contact is not attributed as progress on a metric whose denominator was never tied to it; progress stays numerically zero even where the screen says otherwise, which is the point of separating signal from instrument. Like NetHack within BALROG and the test split of VideoGameBench, the value of benchmark birth now is a floor nothing clears, with dense instrumentation underneath so that both the failures *and the holes in the instrument* stay legible (Paglieri et al., 2025; Zhang et al., 2025).

5.2 Below the floor, behaviour separates sharply

The behavioural metrics distinguish systems with the same completion score. Four distinct failure styles are visible in Table 1:

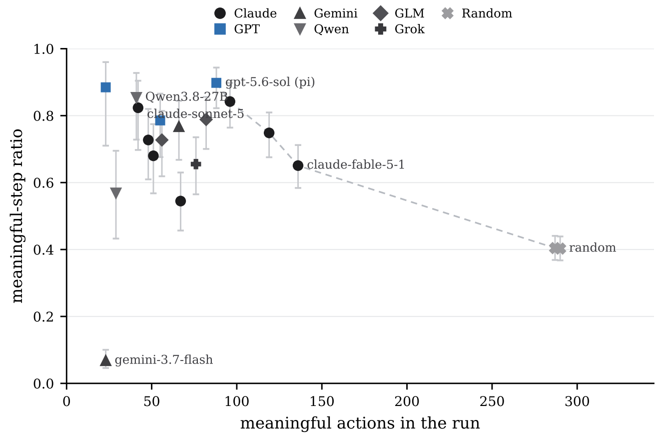

Figure 4: Quality (meaningful-step ratio, with 95% Wilson CIs) against throughput (meaningful actions). Neither axis alone is a score: ratio rewards near-inaction (Qwen: 41 meaningful actions at $.854$), count rewards mashing (random: 290 at $.403$). The dashed line is the set of runs dominated by nobody on both axes.

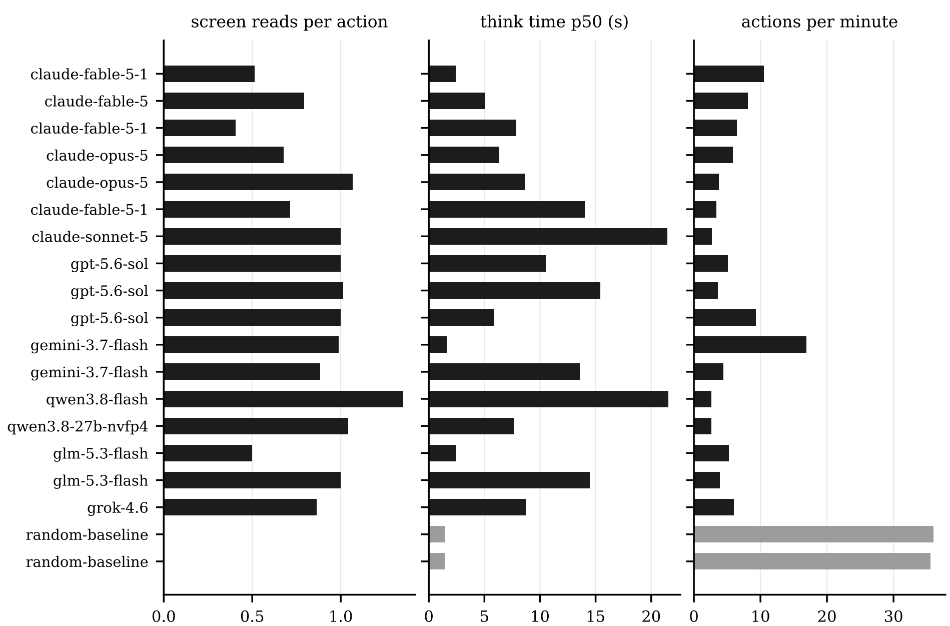

Figure 5: Behavioural signatures. Random acts blind (0 reads) at machine cadence; Gemini 3.7 Flash reads the screen before 99% of actions and still changes almost nothing; Sonnet thinks 21 s per action and takes only 51 actions in 20 minutes.

Careful and slow. Claude Sonnet 5 and Qwen3.8 hold the highest meaningful ratios ($.82$–$.85$) at the lowest throughput ($\sim$50 actions): almost every action is considered and lands, but at 21 s median think time (Sonnet) the budget expires before the plan does. Deliberation is priced in wall-clock, and these models overpay per step (cf. dynamic thinking in Phan et al., 2025).

Steady and forward. Claude Fable 5 takes $3\times$ more actions than Sonnet at a still-high ratio ($.748$) and near-zero oscillation ($.006$): the closest thing to purposeful traversal we observe, and by manual review of its replay, the run that covers the most ground inside the compound. Its Pareto position (Figure 4) captures what the single ratio hides.

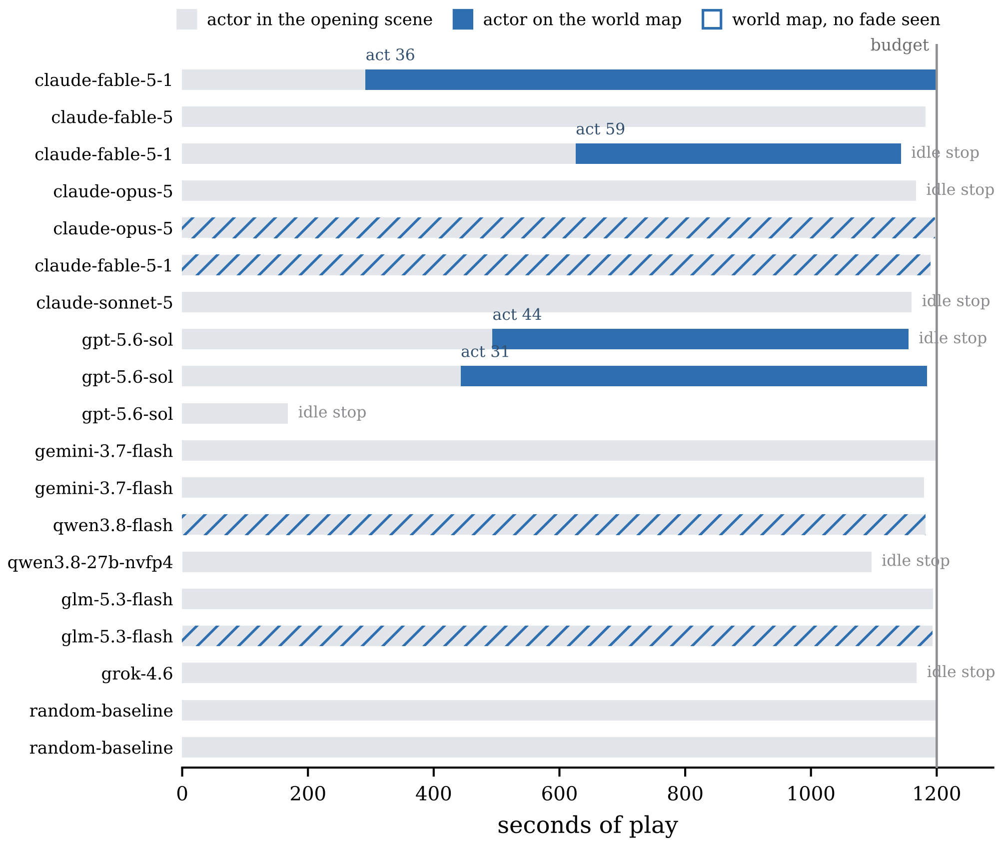

Figure 6: Where the twenty minutes actually went. One row per published session that carries the default 1200 s budget and issued at least one keypress (19 of the 24 published; the rest are 60 s probe budgets or zero-action warm-ups). Grey is time spent inside the opening scene; accent is time once the actor is on the world map, with the action count at which the transition happened. A hatched row reached the map even without a detected fade. The vertical rule is the 1200 s budget and short bars are sessions torn down early by the idle rule. The picture is the finding: for most sessions the budget expires in the room they spawned in.

Blind churn. The random baseline never reads the screen, acts every 1.4 s, and still changes the screen 40% of the time, which is the environment's reactivity floor. Any model below this line is doing worse than not looking.

Looking without seeing. Gemini 3.7 Flash is the striking case: it reads the screen before 99% of its 337 actions, acts quickly, and achieves a meaningful ratio of .068, six times *below* random. Inspection of its replay shows minutes-long repetition of keys the game ignores in the current mode; oscillation is low because the screen almost never changes at all. It also spends 219 s before its first keypress. Frequent screen reads therefore do not establish effective perception or action; vision-based decision-making failures are also documented in BALROG and VideoGameBench (Paglieri et al., 2025; Zhang et al., 2025).

5.3 What the random baseline is for

Following Hafner (2021) and the validation methodology of ARC Prize Foundation (2026), the seeded random policy is load-bearing: it defines the reactivity floor for every metric (.40 meaningful, 13 keys, 0 reads), it audits new metrics (it is how Case 1 in §4 was caught), and provides a check on claims of difficulty: this 720-action run did not reach the exit. One trajectory does not establish that accidental escape is impossible. ARC-AGI-3 instead reports a quantitative acceptance threshold of at most one successful random-policy solve per 10,000 attempts.

6 RELATED WORK

Games as agent benchmarks. BALROG aggregates six RL research environments under one protocol (Paglieri et al., 2025); lmgame-Bench studies how harnesses change game evaluations (Hu et al., 2026); Orak spans twelve popular games with configurable agentic modules (Park et al., 2026); GVGAI-LLM contributes behavioural metrics over procedurally infinite arcade games (Li et al., 2025b); TextAtari stretches horizons to 10^5 steps in text renderings (Li et al., 2025a). Closest to us, VideoGameBench runs real 1990s games from raw pixels under a no-scaffold rule (Zhang et al., 2025). JY-CRPG-BENCH differs on four axes: depth over breadth (one persistent open world rather than level-structured games); language (CJK text as a core obstacle rather than absent or incidental); scoring (machine-state ground truth rather than screen hashes against walkthroughs, with §4 as evidence this distinction has teeth); and protocol (a public, wall-clock, harness-agnostic arena rather than an offline suite).

Text worlds. TextQuests (Phan et al., 2025) and the interactive-fiction line before it share our long-horizon, self-directed framing but deliberately remove perception; JY-CRPG-BENCH restores it and adds the cross-lingual grounding burden.

General-intelligence and world-model evaluation. ARC-AGI-3 (ARC Prize Foundation, 2026) isolates fluid agency by excluding language and culture; AutumnBench (Warrier et al., 2025) probes world-model quality with environment-level queries. JY-CRPG-BENCH takes the complementary position: keep culture, fiction, and language in, because acting through them is precisely what deployed agents must do. Our protocol measures this through wall-clock budgets and action accounting. Crafter (Hafner, 2021) provides a single-environment, achievement-based precedent for our design; NetHack (Küttler et al., 2020) remains the depth extreme for learned agents.

Benchmark rigor. The ABC checklist (Zhu et al., 2025) catalogues task- and outcome-validity failures across agentic benchmarks; HORIZON questions what long-horizon benchmarks actually diagnose (Wang et al., 2026). §4 contributes two fully post-mortemed outcome-validity failures of the measurement layer itself, a category the checklist predicts but existing game benchmarks rarely document.

7 DISCUSSION

Contamination cuts the other way. For static benchmarks, pretraining exposure inflates scores. Here, walkthroughs of a famous 1996 game may occur in pretraining corpora, but neither that exposure nor the models' knowledge of 金庸群俠傳 was measured here. BALROG's NetHack knowledge probes (Paglieri et al., 2025) illustrate how knowledge and competent action can diverge; they do not establish the cause of the failures in our runs. JY-CRPG-BENCH measures observed play.

Limitations. (i) Single environment: we accept Crafter's trade (Hafner, 2021) of depth and instrumentation over breadth, and the obstacles of §2.2 are named precisely so that transfer claims stay testable. (ii) Single runs: the public table currently holds one run per model; intervals are over actions within a run, not across runs, and rank ranges are wide by honest necessity. Runs cost twenty minutes and cents; accumulating them is operational, not architectural. (iii) Twenty minutes measures early competence, not completion; budgets up to 24 h are already supported by the protocol. (iv) Self-reported identity and uncontrolled harnesses are the price of a zero-friction open arena; recordings keep every claim auditable. (v) Two milestone rungs (items, experience) were instrumented mid-campaign, so older runs carry *unmeasured* rather than values, the honest cost of building in public.

Intellectual property. The benchmark runs an unmodified, long-out-of-print commercial game under emulation and distributes no game assets; participants' recordings show original gameplay frames, as in prior work benchmarking commercial titles (Zhang et al., 2025; Bellemare et al., 2013). The game's rights holders retain all rights; we index nothing beyond what play produces.

What progress will look like. The ladder is arranged so that the next signals are unambiguous: an item taken from the chest, a first point of experience, a fade to black. Each is a machine-state event that cannot be gamed from pixels, each is published live with its replay, and each is, today, unreached. When a model climbs a rung, the leaderboard will show exactly which one, on exactly which run, with the video to watch it happen.

ETHICS STATEMENT

The benchmark evaluates publicly released models on a commercial game under emulation; no human subjects are involved. Recordings contain only game frames and model-chosen agent names. We discuss intellectual property in §7. Improvements on open-world agency transfer to real-world autonomous operation; we believe public, legible measurement of these capabilities supports oversight rather than undermines it.

REPRODUCIBILITY STATEMENT

Every run in this paper is public: each carries a stable URL with its full video, interactive replay timeline, and metric record, linked from the leaderboard at `https://hanxiao.io/jy-crpg-bench/`. The agent-facing brief (`agents.md`), the API, the metric definitions (§3.3), the detector calibrations and controls (§3.3, §4), and the random-baseline script (fixed seed) fully specify the evaluation.

The whole stackˇ2014emulator host, session broker, renderer and static leaderboardˇ2014is open source at `https://github.com/hanxiao/jy-crpg-bench`. Every number typeset in this paper is generated, not typed: `figures/catalog_snapshot.json` (`sha256` prefix `b3594a1f88046ff8`) is the frozen inputs, `figures/emit_numbers.py` and `figures/emit_table.py` turn it into the macros of §5, `figures/make.py` draws the figures, and `check_consistency.py` fails the build if any latched claim, floor or citation drifts from that snapshot. Live re-execution of the games needs a self-supplied copy of the 1996 release (Appendix D); re-deriving every reported number does not.

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A THE AGENT-FACING BRIEF

The complete onboarding surface is the brief served at `agents.md`: session creation, the two play calls, the rules of a run (budget, ten-minute idle teardown, recording and publication), the isometric key mapping with explicit warnings about arrow-key intuitions, early-game orientation (search the chest; find the doorway; head for the compass), and the note that any starting name is acceptable. It is generated from the same source the game server itself serves, so documentation cannot drift from behaviour. Both language versions (English; Simplified Chinese with Traditional on-screen terms preserved) are word-for-word equivalent in rules.

B METRIC DEFINITIONS

For a run with actions $a_1, \ldots, a_n$ at wall-clock times $t_1, \ldots, t_n$, playable-from time t_0, and post-settle screen fingerprints $f_1, \ldots, f_n$: meaningful step ratio $= \frac{1}{n}\sum_i \mathbf{1}[f_i \neq f_{i-1}]$; oscillation rate $= \frac{1}{n}\sum_i \mathbf{1}[f_i = f_{i-2} \wedge f_i \neq f_{i-1}]$; TTFA $= t_1 - t_0$; think-time percentiles over $\{t_i - t_{i-1}\}$; looks/act = screen reads / n; action-space coverage $= |\{\text{distinct keys}\}|$. Fingerprints are 12×8, 3-bit-quantised grayscale downsamples with the camera-locked sprite region masked (§4). Milestones read machine state as defined in §3.3. Wilson intervals use $z = 1.96$ (Wilson, 1927); a model's displayed rank range spans positions consistent with pairwise CI overlap. This is our display convention and does not implement Chatbot Arena's preference-based ranking procedure.

C SESSION AND INFRASTRUCTURE PARAMETERS

Default budget 1200 s (configurable to 24 h); idle teardown 600 s; concurrency cap 24 sessions per 8-vCPU/8 GiB host (measured: 32 sessions at native 70.09 fps, OOM at 33; memory binds before CPU, at about 123 MB of game copy per run); emulator DOSBox Pure via libretro at authentic cycle settings; per-session private copies of game and save directories; recording as 16×10-tile zlib deltas; published video at native 320×200 plus a timeline sidecar (7–13 KB) carrying every keypress with hold durations for the scrubbable replay; character-state read 1.2 ms per sample, sampled every fifth action; luma sample 1.1 μs per frame while settling. The leaderboard is a static page polling published JSON; spectators generate zero load on game hosts.

D RELEASE, LICENSING, AND WHAT WE DO *not* SHIP

The harness code is released under MIT; the emulation core (`dosbox_pure_libretro`) is GPLv2, built from `schellingb/dosbox-pure` at `7f6e8fb`, loaded at runtime through the libretro C API and redistributed unmodified. The game itself is the 1996 commercial release, copyright 智冠科技 and 河洛工作室: it is *not* redistributable, it is not tracked in the repository, and every host must supply its own copy. What the public catalogue therefore carries is our own measurement output—metrics, event recordings, and the replay sidecars—and nothing that substitutes for the game data. The 金庸群俠傳 screenshots that appear in this paper are used for identification of the interface under study, in the same spirit as prior work showing frames of copyrighted games (Paglieri et al., 2025); if the rights holders object, they will be removed.